NODAL CURVES ARE GORENSTEIN AND GEOMETRICALLY REDUCED

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file curves.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the lci-Gorenstein chain, the alternative base-change-completion route, the geometric-reducedness reduction, and the reduced normal-form check are all retained verbatim.

Editorial note. The label lemma-facts-about-nodal-curves names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Local complete intersection property of nodal curves. Same chapter, Lemma lemma-nodal-lci; Duality for Schemes, Lemmas duality-lemma-gorenstein-lci and duality-lemma-gorenstein-base-change. Nodal curves are lci, hence Gorenstein; alternatively Gorensteinness can be checked after base change to the algebraic closure.
- Gorensteinness under completion and the local normal forms. Dualizing Complexes, Lemma dualizing-lemma-flat-under-gorenstein; Same chapter, discussion of the completed local rings k[[t]] and k[[x, y]]/(xy). Supports the alternative by-hand verification that the completed local rings of a nodal curve are Gorenstein.
- Geometric reducedness. Varieties, Lemmas varieties-lemma-perfect-reduced and varieties-lemma-geometrically-reduced. Reduces geometric reducedness to reducedness of X_{overline{k}}, where the completed normal forms are visibly reduced.

Source reference. Original lemma 0C03: sources/stacks-chow/tags.txt, source lines 15073–15073. Prerequisite pointer invoked at proof line(s) 4774; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0BVA: sources/stacks-chow/tags.txt, source lines 14903–14903. Prerequisite pointer invoked at proof line(s) 4771; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0BJL: sources/stacks-chow/tags.txt, source lines 14529–14529. Prerequisite pointer invoked at proof line(s) 4777; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

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Source reference. Original reference 0C4E: sources/stacks-curves/pinned-original-source.tex, source lines 4700–4700. Prerequisite pointer invoked at proof line(s) 4770; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 035X: sources/stacks-chow/tags.txt, source lines 4144–4144. Prerequisite pointer invoked at proof line(s) 4785; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 020I: sources/stacks-chow/tags.txt, source lines 2635–2635. Prerequisite pointer invoked at proof line(s) 4784; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let k be a field. Let X be a locally algebraic k-scheme equidimensional of dimension 1 whose singularities are at-worst-nodal. Then X is Gorenstein and geometrically reduced.*

Proof. The Gorenstein assertion follows from Lemma 0C4E and Duality for Schemes, Lemma 0BVA. Or you can use that it suffices to check after passing to the algebraic closure (Duality for Schemes, Lemma 0C03), then use that a Noetherian local ring is Gorenstein if and only if its completion is so (by Dualizing Complexes, Lemma 0BJL), and then prove that the local rings $k[[t]]$ and $k[[x, y]]/(xy)$ are Gorenstein by hand.

To see that X is geometrically reduced, it suffices to show that $X_{\overline{k}}$ is reduced (Varieties, Lemmas 020I and 035X). But $X_{\overline{k}}$ is a nodal curve over an algebraically closed field. Thus the complete local rings of $X_{\overline{k}}$ are isomorphic to either $\overline{k}[[t]]$ or $\overline{k}[[x, y]]/(xy)$ which are reduced as desired. □